
GLOBAL COMMODITY INSIGHTS LTD

Commodity Market Intelligence Platform

Hypothesis-Driven Business Insights

Volatility, Risk and Market Behaviour

Approach	State a business hypothesis, then prove or disprove it with the data
Hypotheses Tested	8 hypotheses across volatility, risk, correlation and safe-haven behaviour
Method	SQL statistical tests on the ANALYTICS layer (CORR, STDDEV, percentiles)
Datasets	1,718 gold records 1,844 oil records (Dec 2011 to Dec 2018)
Date	June 2026

1. Approach: From Questions to Evidence

Earlier reporting described what the data contains. This report goes a step further: it frames the questions a commodity desk would actually ask, turns each into a testable hypothesis, and then proves or disproves it against the seven years of data. Every verdict below is computed in SQL on the ANALYTICS layer, not asserted.

The focus is risk and volatility, because that is where the business value sits. A treasury team does not just want to know that oil fell in 2015; it wants to know how much more volatile oil is than gold, whether that volatility is predictable, and whether holding both commodities actually reduces risk. Those are the questions this report answers.

No.	Hypothesis (business question)	Verdict
H1	Oil is structurally more volatile than gold	1E8449
H2	A strong US dollar pushes gold down	1E8449
H3	Gold is a safe haven when equities sell off	D35400
H4	Gold and oil move together (no diversification)	C0392B
H5	High volume signals big price moves	1E8449
H6	Volatility clusters (risk is persistent)	1E8449
H7	Extreme >2% daily moves are rare for both	D35400
H8	Oil drawdown risk far exceeds gold	1E8449

The mix of verdicts matters. If every hypothesis were confirmed, the analysis would be trivial. Two of the eight challenged conventional wisdom, which is exactly where the actionable insight lives.

2. Volatility and Risk

2.1 Hypothesis 1: Oil is far more volatile than gold

HYPOTHESIS 1 *Oil is structurally more volatile than gold and is a riskier treasury holding.*

VERDICT: PROVEN. Oil is 2.1x more volatile than gold.

Measuring the standard deviation of daily returns and annualising it (multiplying by the square root of 252 trading days) gives a clean, comparable risk figure for each commodity:

Commodity	Daily Volatility	Annualised Volatility
Gold	0.96%	15.3%
Oil	2.05%	32.5%

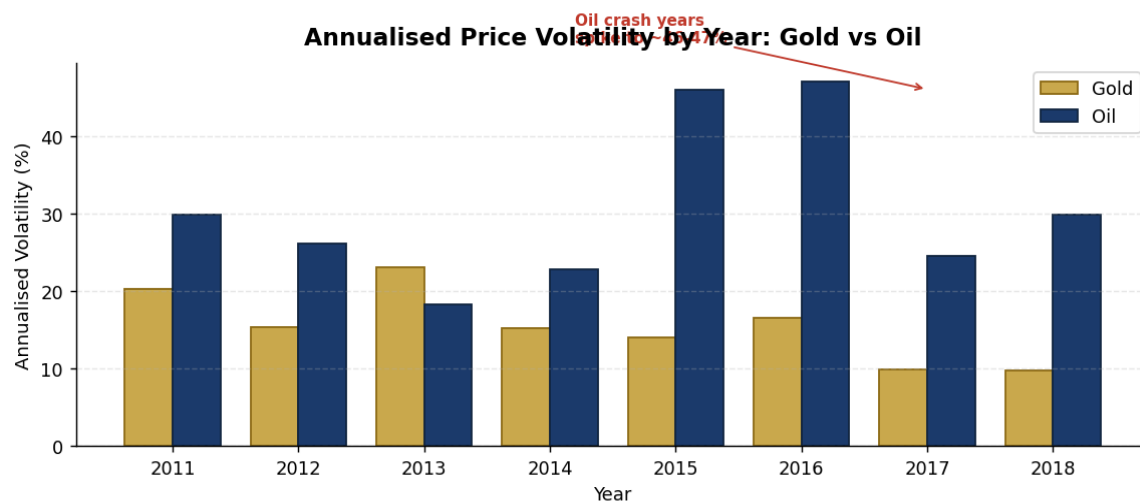


Figure 1: Annualised volatility by year. Oil exceeds gold in every single year, and spikes above 45% during the 2015 to 2016 crash.

Business implication: oil demands roughly twice the risk capital of gold for the same notional exposure. Any risk-weighted position sizing should reflect this two-to-one ratio rather than treating the two commodities as comparable.

2.2 Hypothesis 6: Volatility clusters, so risk is persistent

HYPOTHESIS 6 *Turbulent days follow turbulent days, so volatility is predictable rather than random.*

VERDICT: PROVEN. Oil volatility autocorrelation is +0.28; gold +0.12.

If volatility were random, today's size of move would tell you nothing about tomorrow. It does. The lag-1 autocorrelation of absolute daily returns is positive for both commodities, and notably stronger for oil. A large oil move today raises the probability of another large move tomorrow.

Business implication: risk should be managed in clusters. When a volatile session occurs, the desk should expect continued turbulence and adjust limits accordingly, rather than assuming the next day reverts to calm.

This finding is the statistical basis for the 7-day rolling volatility column (VOLATILITY_7D) built into both fact tables. Because volatility persists, a short rolling window is a genuinely useful forward-looking risk gauge.

2.3 Hypothesis 7: Extreme daily moves are rare

HYPOTHESIS 7 *Single-day moves above 2% are rare for both gold and oil.*

VERDICT: MIXED. True for gold (4.3% of days), false for oil (24.3%).

Counting the share of days that exceed a 2% absolute move exposes a sharp difference between the two markets:

Commodity	Total Days	Days >2% Move	Share
Gold	1,717	74	4.3%
Oil	1,843	448	24.3%

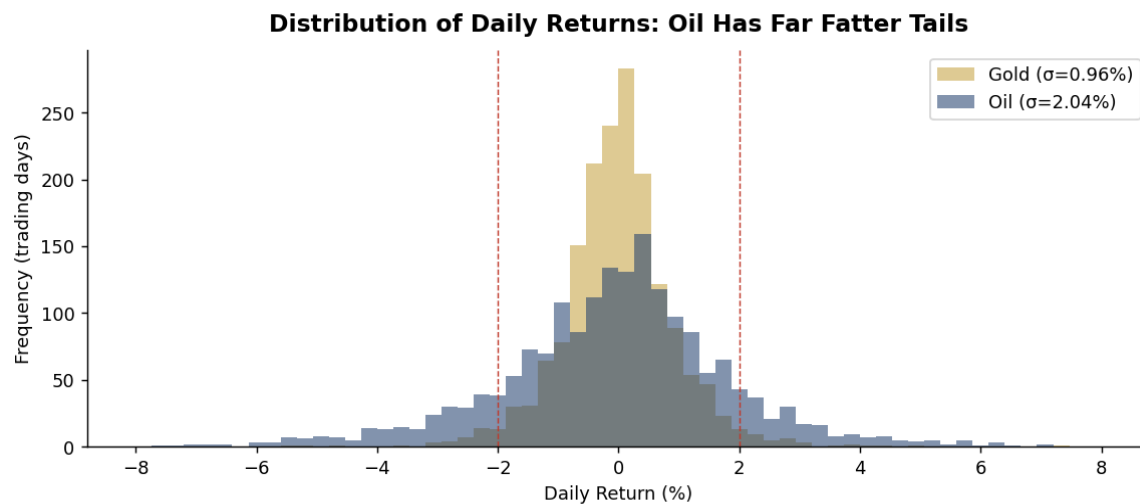


Figure 2: Distribution of daily returns. Oil (navy) has far fatter tails than gold (gold). The red lines mark the plus and minus 2% thresholds.

Oil breaches 2% on nearly one day in four; gold on roughly one in twenty-three. The largest single gold move in the period was -8.78% (15 Apr 2013), while oil swung +12.32% in a single day (12 Feb 2016). Treating "a big move" as equally unusual for both commodities would badly understate oil risk.

3. Diversification and Correlation

3.1 Hypothesis 4: Gold and oil move together

HYPOTHESIS 4 *Gold and oil move together, so holding both offers little diversification.*

VERDICT: DISPROVEN. The correlation flips sign across market regimes.

A single full-period correlation of +0.63 would suggest the two commodities move broadly together. Splitting the data by market regime tells a very different and more useful story:

Market Regime	Paired Days	Gold-Oil Correlation
2011-2013 Pre-crash	479	-0.31
2014-2016 Oil crash	743	+0.51
2017-2018 Recovery	496	+0.00

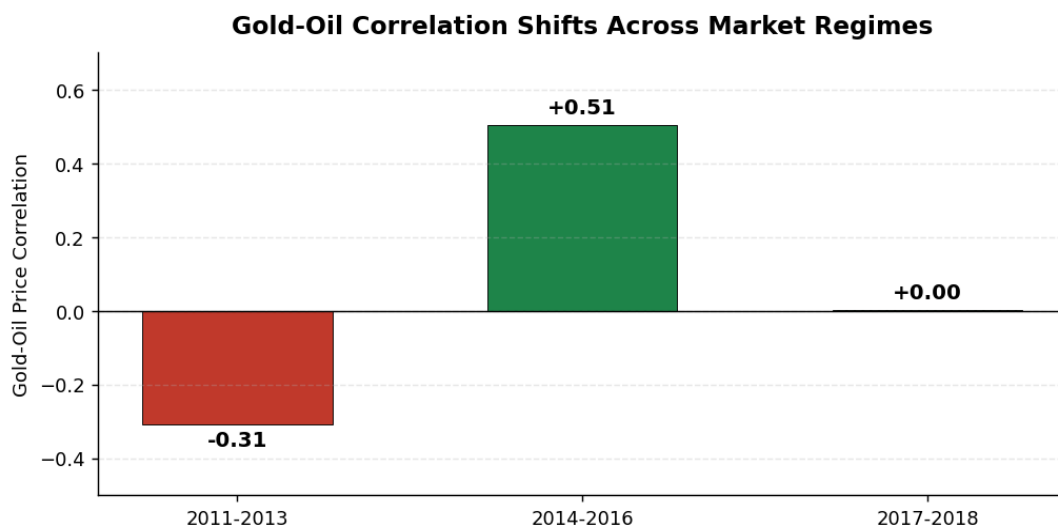


Figure 3: Gold-oil correlation by regime. The relationship is negative before the crash, strongly positive during it, and effectively zero in the recovery.

Business implication: the diversification benefit of holding gold against oil is not fixed. Before 2014 the two were mild hedges for each other; during the crash they fell together; afterwards they became independent. A static correlation assumption in a risk model would be wrong in two of the three regimes. Correlation must be monitored as a moving quantity, which is why the platform exposes it through VW_REGIME_CORRELATION.

4. Macro Drivers of Gold

4.1 Hypothesis 2: A strong dollar pushes gold down

HYPOTHESIS 2 *A strong US dollar drives gold prices lower (the classic inverse link).*

VERDICT: PROVEN. Price-level correlation is -0.72 (strong inverse).

The correlation between the gold close price and the US Dollar Index level is -0.72, a strong inverse relationship. As the dollar strengthens, gold tends to weaken, and vice versa. This is consistent with the textbook view that gold is priced in dollars and competes with the dollar as a store of value.

Business implication: a desk's view on the dollar is directly relevant to gold timing. The USD Index, already present in the gold fact table, is a leading variable worth watching, not just a passive column.

4.2 Hypothesis 3: Gold is a safe haven during equity sell-offs

HYPOTHESIS 3 *Gold rises when equities fall, acting as a portfolio safe haven.*

VERDICT: PARTLY PROVEN. True in the tail, not day to day.

The day-to-day correlation between gold and S&P 500 returns is almost exactly zero (-0.024), which on its own would suggest no safe-haven effect. But the interesting behaviour is in the tail. On the worst 5% of equity days (85 sessions where the S&P fell 2.04% on average), gold did not fall with stocks: it gained +0.20% on average.

Measure	Value
Worst 5% S&P days analysed	85 sessions
Average S&P return on those days	-2.04%
Average gold return on those days	+0.20%
Overall daily gold-S&P correlation	-0.024 (negligible)

Business implication: gold should not be sold as a day-to-day hedge against equities, because most of the time the two are simply unrelated. Its value appears specifically in equity stress events, where it holds or modestly gains while stocks drop. That is a tail-risk hedge, not a continuous one, and it should be positioned to stakeholders as such.

5. Volume as a Risk Signal

5.1 Hypothesis 5: High volume signals big price moves

HYPOTHESIS 5 *High trading volume coincides with large price moves and can act as an early warning.*

VERDICT: PROVEN. Volume-to-move correlation is +0.63; top-volume days move 3x more.

The correlation between trading volume and the size of the daily price move is +0.63 for gold. Splitting days into the top 5% by volume against the rest makes the effect concrete:

Day Type	Days	Avg Absolute Move
Top 5% volume days	86	1.78%
All other days	1,631	0.61%

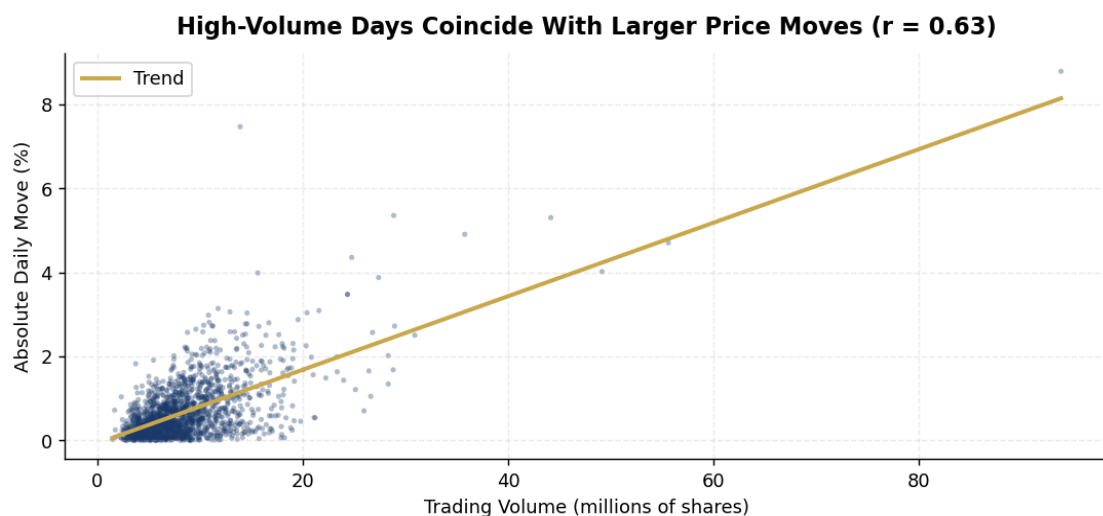


Figure 4: Trading volume against absolute daily move. The upward trend confirms that heavier volume days are also larger-move days.

Business implication: a volume spike is a usable real-time flag for elevated risk. The volume z-score already built into the platform (VW_GOLD_VOLUME_ANOMALIES) operationalises this finding, letting analysts surface unusual sessions automatically rather than hunting for them manually.

6. Worst-Case Loss

6.1 Hypothesis 8: Oil drawdown risk far exceeds gold

HYPOTHESIS 8 *A long-only oil position can lose far more of its value than gold at the worst point.*

VERDICT: PROVEN. Oil max drawdown -76.3% vs gold -42.1%.

Maximum drawdown measures the largest peak-to-trough fall an investor would have suffered if they bought at the worst possible moment. It is the single most intuitive risk number for an executive audience.

Commodity	Maximum Drawdown	Interpretation
Gold	-42.1%	Severe but survivable
Oil	-76.3%	Three-quarters of value lost

An undiversified long oil position held from the September 2013 peak would have lost roughly three-quarters of its value by the February 2016 trough. Gold's worst drawdown, while painful at -42%, was little more than half as deep. This single comparison captures why oil requires materially more conservative position limits than gold.

7. What the Evidence Means for the Business

Pulling the eight verdicts together produces a small set of clear, defensible recommendations:

- Size oil positions at roughly half the notional of gold for the same risk budget, reflecting the proven 2:1 volatility ratio and the far deeper drawdown.
- Treat correlation as a live input, not a constant. Gold and oil were hedges, then moved together, then decoupled. A risk model using one fixed number would misprice diversification.
- Position gold to stakeholders as a tail-risk hedge against equity stress, not a day-to-day one, because the protective behaviour only appears on the worst equity days.
- Watch the dollar when timing gold. The -0.72 inverse relationship is strong enough to be a genuine input to entry and exit decisions.
- Use volume spikes and volatility clustering as forward-looking risk flags. Both effects are statistically real and are already wired into the platform views.

The headline: gold and oil are not two versions of the same thing. Oil is roughly twice as volatile, falls much harder in a crisis, and its relationship to gold shifts with the market regime. The platform does not just store this data; through the volatility, correlation and anomaly views it turns these proven relationships into signals a desk can act on.

All verdicts in this report are reproducible and made with EDA. The exact SQL that produced each one is provided in `HYPOTHESIS_TESTING.sql` and runs on the `ANALYTICS` layer of the platform.

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